The Revolution Beneath the Revolution

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Today's AI revolution rests on a deeper revolution:

the transformation of proofs into mathematical and computational objects.

The history of modern AI is, in part, the history of proof theory.

Everyone is talking about the AI revolution.

I want to talk about an older revolution.
underneath it.

AI did not emerge from nowhere.

It inherited a century of ideas about representing, manipulating, and automating **reasoning**.

Frank Quinn (AMS Notices, 2012):

> *"A revolution in mathematics? What really happened a century ago and why it matters today."*

Main claim:

- Mathematics experienced a genuine revolution.
- It changed the nature of mathematical practice.
- Yet many mathematicians barely noticed it.

RIGOUR

Mathematics becomes formal.

$\Downarrow$

STRUCTURE

Proofs become objects.

$\Downarrow$

SCALE

Machines reason with millions of proofs.

Hilbert $\rightarrow$ Curry–Howard $\rightarrow$ AI

1890–1930

- Frege
- Hilbert
- Russell
- Gödel
- Gentzen
- Turing

Proofs become mathematical objects.

The revolution was not about new theorems.

It was about new questions.

- What is a proof?
- What counts as rigour?
- Can mathematical reasoning itself be studied mathematically?

Formalise mathematics.

Prove its consistency.

Understand mathematical reasoning itself.

1931

> *Any sufficiently expressive formal system is either incomplete or inconsistent.*

Hilbert's original programme could not succeed.

Yet the revolution survived.

Proofs had become mathematical objects.

Church

λ-calculus

Turing

machines

Computation becomes a mathematical subject.

What if proofs are computations?

Proofs $\leftrightarrow$ Programs

Propositions $\leftrightarrow$ Types

Not a theorem.

A research programme.

The correspondence revealed that

- proofs have computational content;
- proof normalisation resembles computation;
- logical systems can be interpreted computationally.

A proof is more than a certificate of truth.

The Revolution of Structure

Proofs are not only objects.

They have structure.

They compose.

They can be transformed.

Category Theory Enters

Lambek, Lawvere

Hyland, and many others

Proofs as morphisms

Composition as a first-class concept

Semantics as mathematics

Not because it is abstract.

Because it organises.

- logic and computation;
- syntax and semantics;
- proofs and programs;
- different mathematical disciplines.

A mechanism for mathematical self-organisation.

Because it changed

- what proofs are;
- what programs are;
- how logic is interpreted;
- how different disciplines communicate.

And because the consequences are still unfolding.

Martin-Löf Linear Logic HoTT

Proof Assistants Formal Verification

The revolution of rigour succeeded.

The revolution of structure is still happening.

We do not yet know:

- its final form;
- its limits;
- its ultimate applications.

AI is exposing how incomplete our understanding of reasoning still is.

If proofs are mathematical objects,

then we still need to understand:

- their semantics;
- their composition;
- their transformations;
- their identity criteria;
- their interaction with computation;
- their interaction with learning.

These questions are still largely open.

The Curry–Howard correspondence tells us

proofs are programs.

But a correspondence is not enough.

We need **mathematical models** that explain:

- 'when proofs compose',
- 'why programs compute',
- 'which transformations preserve meaning',
- etc...

This is the role of semantics.

Gödel introduced Dialectica (1958) as an interpretation of arithmetic.
Later it became:

- a semantic construction,
- a method for extracting constructive content of proofs,
- a bridge between proof theory and category theory,
- several bridges between logic and computation,
- a way of transporting logical ideas between systems.

A precursor of many later developments.

One question has guided much of my research.

What can we learn if we treat proofs as mathematical objects?

This led me to explore Dialectica categories as semantic models for

- linear logic,
- game semantics,
- Petri nets,
- programming-language semantics,
- imperative type systems,
- small cardinals,
- and, more recently, AI for mathematics.

Different applications, one research programme.

Before Gentzen: proofs certify truth.

After Gentzen: proofs become mathematical objects.

My work: understanding the mathematics of those objects.

What happens when those objects are manipulated not by mathematicians, but by machines?

AI

Proof Assistants

Formal Verification

Mathematics at scales beyond direct human inspection.

Generating ‘mathematics’ is now easy.

Verifying mathematics at scale is hard.

This is fundamentally a semantic problem.

Rigour (Hilbert, Gödel, Gentzen)

$$\Downarrow$$

Structure (Curry–Howard, Lambek, Lawvere)

$$\Downarrow$$

Scale (AI, Formalisation, Verification)

The Curry–Howard revolution is still happening.

Rigour (Hilbert, Gödel, Gentzen)

$$\Downarrow$$

Structure (Curry–Howard, Lambek, Lawvere)

$$\Downarrow$$

Scale (AI, Formalisation, Verification)

The Curry–Howard revolution is still happening.

The AI revolution is just starting

Mathematical revolutions are not sustained by ideas alone.

They need

theories,
tools,
people,
communities.

Only then do they become part of mathematics.

The first revolution gave us rigour.

The second gave us structure.

The third is giving us scale.

What will determine what comes next?

Not just logical systems.
Not just algorithms.
Not just hardware.
But the people who choose the questions worth asking.

Ideas do not spread by themselves.

Research programmes need

people,
collaboration,
shared standards,
places to meet,
communities.

Women in Logic is our community

Women in Logic (WiL) is a workshop series co-located with major logic conferences.

- Founded in 2017 at LICS in Reykjavík.
- Annual since then — through pandemics, hybrid formats, nine cities.
- Website: `https://womeninlogic.org/`

The name was chosen deliberately.

The internet had already decided what "Women Logic" meant.

Search "Women in Logic" online and you find
a long-standing sexist meme.

The premise: *women are not logical.* Ha ha.

In the early years of WiL, googling the workshop
would first surface, without apparent irony:

> *Edward E. Hale, Women and Logic,*
> *The North American Review,*
> *Vol. 198, No. 693 (Aug. 1913), pp. 206–217.*

> *"That women are not logical is one of the recognized conventions of social life."*

The internet had decided what "Women" + "Logic" meant.
We disagreed.

Nine Editions, Nine Venues

Year	Edition	Co-located with
2017	1st WiL	LICS, Reykjavík
2018	2nd WiL	FLoC, Oxford
2019	3rd WiL	LICS, Vancouver
2020	4th WiL	FSCD, Paris / online
2021	5th WiL	LICS, Rome / online
2022	6th WiL	FLoC, Haifa
2023	7th WiL	FSCD+CADE, Rome
2024	8th WiL	LICS/ICALP/FSCD, Tallinn
2025	9th WiL	Birmingham

120 talks over nine years.

Recently compiled at a spreadsheet Women_in_Logic_10years in our website.

106 distinct speakers.

19 invited talks · 101 contributed talks.

12 speakers returned for more than one edition.

Sara Uckelman and Liron Cohen each spoke *three* times.

A community developing, not just a list.

- Automated Reasoning & Verification 18
- Modal & Non-Classical Logic 18
- Type Theory & PL 17
- Computability & Complexity 16
- Proof Theory 15
- Automata & Formal Languages 10
- AI & Applications 8
- Philosophy & History 8
- Categorical & Algebraic Logic 4
- Natural Language 3
- Probabilistic Methods 3

The full breadth of logic is here.

Some speakers came back — and back again:

Speaker	**Editions**
Sara L. Uckelman	3 times
Liron Cohen	3 times
Sandra Kiefer	2 times
Shufang Zhu	2 times
Delia Kesner	2 times
Giselle Reis	2 times
Manon Blanc	2 times
Ilina Stoilkovska	2 times
Iris van der Giessen	2 times

Return speakers signal that WiL is a *home*, not just a venue.

WiL is more than an annual event. It is a collective of like-minded people.

Our Steering Committee consists of:

- Agata Ciabattoni
- Alexandra Silva
- Amy Felty
- Brigitte Pienka
- Sandra Alves
- Sandra Kiefer
- Sara L. Uckelman
- Sonja Smets
- Valeria de Paiva

- **Website:** `https://womeninlogic.org/`
  full record of all editions, speakers, proceedings
- **WiL ONLINE** `https://womeninlogic.org/wil-online/`
  Organized by Elaine Pimentel, Maria Osório Costa and Ekaterina Piotrovskaya at UCL. Previously organized by Agata Ciabattoni and Josephine Dik.
- **Facebook group:** Women in Logic
  ongoing community, announcements, support
- **Blog:** Women in Logic `https://blog.womeninlogic.org/`
- **Social Media:** Bluesky, LinkedIn and Instagram coordinated by different people
- **Mailing list:** Sandra Kiefer moderates, connecting women in logic year-round

Visibility is not automatic. It requires maintenance.

Mathematical communities shape

- what gets funded;
- what gets cited;
- who gets invited.

Invisible contributions stay invisible.
Invisible researchers stay invisible.

Making the record visible
is itself a form of mathematical work.

More than 106 women doing serious logic research, 2017–2025.

Across proof theory, type theory, model theory, verification, automata, AI.

From institutions on every continent.

The talent was always there.

The question is always: who gets to be seen?

Formal logic made mathematics rigorous.
Proof theory is making reasoning manipulable.
AI is making reasoning scalable.

The revolution beneath the AI revolution is not over.
We're part of it.

WiL is ten years old.
The field is older.

The work continues.

With thanks to all WiL organisers, presenters, PC memebers and attendees past and present.

Especial thanks to

Elaine Pimentel, Anela Lolic and Huimin Dong
(this year's organizers)
and also especially to **Tim Hosgood** and **Evelyn Erickson**

(for their work behind the scenes).

https://womeninlogic.org/

Facebook: Women in Logic